

Fast and Accurate Camera Scene Detection on Smartphones

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Abstract

AI-powered automatic camera scene detection mode is nowadays available in nearly any modern smartphone, though the problem of accurate scene prediction has not yet been addressed by the research community. This paper for the first time carefully defines this problem and proposes a novel Camera Scene Detection Dataset (CamSDD) containing more than 11K manually crawled images belonging to 30 different scene categories. We propose an efficient and NPU-friendly CNN model for this task that demonstrates a top-3 accuracy of 99.5% on this dataset and achieves more than 200 FPS on the recent mobile SoCs. An additional in-the-wild evaluation of the obtained solution is performed to analyze its performance and limitation in the real-world scenarios. The dataset and pre-trained models used in this paper are available on the project website.

1. Introduction

Camera scene detection is one of the most popular computer vision problems related to mobile devices. Nokia N90 released in 2005 was the world’s first smartphone with a manual camera scene selection option containing five categories (close-up, portrait, landscape, sport, night) and different lighting conditions (sunny, cloudy, incandescent, fluorescent) [36]. Notably, it was also able to select the most appropriate scene automatically, though only basic algorithms were used for this and the result was not always flawless. Since then, this became a standard functionality for the majority of camera phones: it is applied to accurately adjust the photo processing parameters and camera settings such as exposure time, ISO sensitivity or white balancing to get the best image quality for various different scenes. For instance, certain situations require a high shutter speed to avoid the picture being blurry. A good example of this are

pictures of animals, sport events or even kids. A modified tone mapping function is often needed for portrait photos to get a natural skin color, while special ISO sensitivity levels are necessary for low-light and night photography. An appropriate white balancing method should be used for indoor photos with artificial lighting so that the resulting images have correct colors. Finally, macro and portrait photos are often shot using bokeh mode [17] that should be enabled automatically for these scenes. Therefore, the importance of the camera scene detection task cannot be underestimated as it drastically affects the resulting image quality.

Using the automatic scene detection mode in smartphone cameras is very easy and convenient for the end user, but this poses the problem of making accurate predictions. The first scene classification methods were based on different heuristics and very simple machine learning-based algorithms as even the high-end mobile devices had at best a single-core 600 MHz Arm CPU at that time. The situation changed later when portable devices started to get powerful GPUs, NPUs and DSPs suitable for large and accurate deep learning models [21, 19]. Since then, various AI-powered scene detection algorithms appeared in the majority of mobile devices from Huawei [9], Samsung [24], Xiaomi [43], Asus [1] and other vendors. However, since no available public datasets and models were available for this task, each manufacturer was designing its own solution that was often capable to recognize only a very limited number of classes.

To address the above problem, in this paper we present a novel large-scale *CamSDD* dataset containing more than 11 thousand images and consisting of the 30 most important scene categories selected by analyzing the existing commercial solutions. We propose several efficient MobileNet-based models for the considered task that are able to achieve a top-1 / top-3 accuracy of more than 94% and 99%, respectively, and can run at over 200 FPS on modern smartphones. Finally, we perform a thorough performance evaluation of the proposed solution on smartphones in-the-wild and test its predictions for numerous real-world scenes.

The rest of the paper is arranged as follows. Section 2 reviews the existing works related to image classification

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